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America's continued dominance in the AI space depends not just on computer screens and coding, but on those who work with their hands. It is in that spirit that Gecko Robotics respectfully submits the enclosed proposals to the White House Office of Science and Technology Policy as the Administration weighs a new AI Action Plan to maintain and advance America's critical place as the global leader in AI.

Gecko Robotics is an AI and robotics company transforming how the world builds, operates, and maintains its most critical infrastructure. At Gecko we believe that combining first-order data collected from infrastructure with the predictive power of AI can create a more reliable and resilient future for the physical world. Gecko agrees with the Administration that America must solidify its position as the global leader in AI to secure a prosperous future for all Americans, and we can do this by bolstering two critical elements of the AI Supply Chain: energy and data.

Vice President Vance made an important point during his speech at the AI Summit in Paris last month when he stated, "now it's my view that tech innovation over the last 20 years has often conjured images of smart people staring at computer screens, engineering in the world of bits, but the AI economy will primarily depend on and transform the world of atoms." If the entire focus of American AI initiatives is, "*smart people staring at computer screens*" we will lose.

Energy remains a critical input for accelerating and expanding the development of AI for the US. As we develop and expand the role of AI in our everyday lives, the substantial computational power of AI will translate into increasingly significant energy requirements. Efficiently scaling AI solutions directly depends on our infrastructure's capacity to reliably generate and distribute this power. By investing in reliable and resilient energy infrastructure, the U.S. can ensure that we meet current and future AI needs. This investment is critical for creating a robust foundation for AI-enabled innovation.

Data is the other foundational input necessary to realize the transformative potential of AI. High-quality, high-volume data is the fuel that powers advanced AI models, enabling them to learn, adapt, and make intelligent decisions. Ensuring access to comprehensive, accurate, first-order data will drive the effectiveness of AI applications across our most critical economic sectors. High-quality data enhances model precision, improves outcomes, and supports the expansion of AI's role in boosting productivity and economic growth. However, to date, securing this high-quality data has remained a persistent challenge, especially for data on physical infrastructure. There is very little high-quality data collected on the physical structures we rely upon, from bridges to factories to ships.

By focusing on energy and data, two critical foundations for AI development and enablement, the U.S. can ensure that it not only remains at the forefront of AI innovation, but also leverages these advancements to foster economic prosperity and improve the quality of life for the American people.

1) Create a Technology-Forward National Strategy for Optimizing Power Generation to Meet Energy Needs

America is in an energy crisis. And the growing strain to keep up with the demands of AI are an increasingly important factor. AI's computational power is projected to [double every 100 days](#).¹ In part due to the strain of AI on data centers, experts project an increased national energy demand of around 16% over the next four years, adding greater strain on an already fragile U.S. grid.

What America does best is innovate. American history is steeped in stories of innovation leading us out of a crisis. The transistor helped America win the space race. Radar and jeeps helped win World War II. And American technology centers are the wonder of the world. Innovation has always been the cornerstone of how America responds to a crisis. And - as President Trump declared in January - we are currently in an energy crisis. Despite the importance, and scale, of the problem we are not deploying the power of American innovation across our energy infrastructure.

America will likely lose the AI race if we cannot generate the energy necessary to meet its increasing demands. To support the growing demand, it is imperative that the U.S. harnesses advanced technologies to optimize power generation across all energy sources - from fossil fuels to nuclear and renewables. By retrofitting and supporting our aging power infrastructure with cutting-edge technology such as robotics, automation, predictive maintenance and real-time analytics, we can put another example of America using innovation to come out ahead during a crisis.

Proposal: The Department of Energy (DOE) should encourage the use of advanced technology across all sources of energy production to optimize their operations with the goal of increasing production and reducing downtime.

The generation facilities of today are the assets powering the AI of tomorrow. To bolster American AI dominance, the U.S. should establish a national strategy for optimizing the operation of all sources of energy that power our grid. That should include both optimizing currently operating facilities and reducing downtime for facilities currently offline. To do so, DOE should issue regulations that would require power production facilities to use advanced technology like robotics and automation to ensure the power grid is resilient and reliable in the face of the growing demand for energy. These technologies will drastically reduce downtime of power facilities while increasing reliability and production. DOE should also make loan and grant funding available for power production facilities to optimize their operations with advanced technology.

¹ New Report Analyzes Potential for AI to Fuel Major Increase in Power Demand. American Public Power Assoc. June 20, 2024. <https://www.publicpower.org/periodical/article/new-report-analyzes-potential-ai-fuel-major-increase-power-demand>

The electricity grid consists largely of three stages: generation, transmission, and distribution. In recent years, transmission and distribution have received a lot of attention and funding. This has been driven by the need to connect wind and solar power (often generated far away from population centers) to the larger grid to move it into more densely populated areas. However, data centers are discrete, power-hungry, facilities that are [increasingly being built close to generating facilities](#) themselves.² These “behind the meter” connections do not require the same transmission and distribution investments that are needed to power residential centers throughout the country. To power AI, the portion of the grid that now needs more attention and investment is generation.

The Administration’s AI Action Plan should recognize that the relationship between establishing AI dominance and energy dominance are mutually beneficial: AI designed for the built world can ensure that all other AI applications have adequate power – while ensuring adequate power for American AI companies will foster the next generation of advancements in AI. In fact, Gecko is now deploying AI and robotics alongside the United States’ largest independent power plant operator to identify potential dangers and optimization points at energy generating facilities.³

2) Require AI-ready Data for Every New Build for Government Infrastructure

As AI continues to develop, the field will see a proliferation of open-source base models for AI applications. AI giants like NVIDIA have [released several open-source](#) models for digital twins and physical AI systems (like OpenUSD, Omniverse, etc.).⁴ As these models become widely available throughout the industry, the differentiator that will set American AI companies apart from their competitors will be the quality of data they have access to.

The large language model (LLM) craze of recent years was fed largely by data that existed on the Internet and was repurposed for training the ChatGPTs and Claudes of the world. Now, anyone can spin up an LLM with a few lines of code and an Internet connection. For AI applications in service-based sectors like finance and eCommerce, models have been trained on data generated over decades of transactions.

But for the physical world—in industries ranging from energy and public infrastructure to manufacturing and transportation—we do not have anywhere near enough data to enable AI

² Big Tech wants to plug data centers right into power plants. Utilities say it’s not fair. AP News. Jan 25, 2025. <https://apnews.com/article/power-electricity-amazon-microsoft-power-plants-data-centers-grid-f4763f73bc112425e18f30618dff0039>

³ Gecko Robotics announces \$100M deal with NAES. Pittsburgh Business Times. Feb 27, 2025. <https://www.bizjournals.com/pittsburgh/news/2025/02/27/gecko-robotics-announces-100m-deal-with-naes.html?>

⁴ The Era of AI-Enabled Digital Twins Has Arrived. NVIDIA. Accessed March 2025. <https://www.nvidia.com/en-us/omniverse/solutions/digital-twins/>

solutions. In order for the U.S. to win the AI arms race, we need to have better data on everything that matters—and that includes infrastructure.

Proposal:

Since critical infrastructure is largely managed by public entities, collecting the next generation of data to power physical AI will require both permission and investment from government agencies. Efforts to accelerate the development of physical AI systems include:

(A) When the government funds a major infrastructure project, agencies should require a high-fidelity digital twin created at the time of construction.

This digital twin should be informed by high-fidelity data on the physical condition of the asset to ensure that “as-built” assets match the “as-designed” schematics. In the same way that consumers are offered manufacture warranties on physical goods, agencies should procure high-fidelity digital twins at the time of construction to ensure their assets are built as intended and to reduce long-term lifecycle costs. Collecting high-quality data over the lifetime of an asset will allow for targeted repairs, less downtime, fewer catastrophic accidents, and savings of billions of dollars.

(B) Require all capital-intensive assets critical for U.S. national security and defense to have a full digital scan of their physical condition.

The impact of this recommendation is two-fold. First, it would allow trusted AI companies to begin developing critical datasets on how capital-intensive assets respond throughout their lifecycle to changes in operational conditions. By analogy, LLM’s were only possible because decades of text-based data were available to developers. In many instances, this data is not yet collected in a machine-readable way for critical security applications.

Second, it would create a permanent auditable “digital thread” of the condition of capital-intensive assets from the time of creation throughout its entire life cycle. This has important implications for QA/QC of major acquisition programs.

Consider the process of inspecting welds on large assets like submarines and aircraft carriers. Currently, weld inspections are largely a manual process. A weld inspector uses the appropriate instrument to evaluate raw data coming from the weld and records their assessment on the quality of the weld. However, when robotics assist with weld inspections, the raw data can be permanently recorded - alongside the inspectors’ expert assessment of weld material. Given that the Navy has recurring struggles ensuring the adequacy of contractors’ welds (including in [2007](#), [2009](#), and most recently in [2024](#)), creation of this permanent “ground-truth” dataset could accelerate the identification

of faulty welds.^{5,6} This in turn could accelerate timelines for the production of critical assets to meet national security priorities while reducing the possibility of serious safety incidents.

We welcome the opportunity to discuss these proposals with the Office of Science Technology and Policy and thank you for your consideration.

⁵ More welding woes uncovered for Newport News shipyard. The Virginian-Pilot. June 8, 2009.
<https://www.pilotonline.com/2009/06/08/more-welding-woes-uncovered-for-newport-news-shipyard/>

⁶ Lawmakers to Investigate Faulty Sub, Carrier Welding at Newport News Shipbuilding. USNI News. Sep 27, 2024.
<https://news.usni.org/2024/09/27/lawmakers-announce-investigation-into-faulty-submarine-carrier-welding-at-newport-news-shipbuilding-ships-affected-in-low-single-digits-officials-say>